

Module/Pathway/Taxon Review: tRNA ms2io6A37 Hydroxylation (UPA00729) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (organism code PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** UniPathway UPA00729 — terminal O₂-dependent hydroxylation of ms2i6A37-tRNA → ms2io6A37-tRNA **Candidate gene(s):** miaE (PP_2188 | UniProtKB Q88KV1)

1. Executive Summary

The UniPathway UPA00729 bucket for *P. putida* KT2440 contains a single enzyme member, **MiaE (PP_2188, Q88KV1)**, the tRNA (ms2io6A37) hydroxylase. On the narrow question of whether that terminal hydroxylation *enzyme* is present and functional in KT2440, the evidence is **unusually strong**: Q88KV1 is a reviewed Swiss-Prot entry with protein-level existence evidence (PE1), four experimental X-ray structures (PDB 2ITB, 6ZMA, 6ZMB, 6ZMC), an assigned EC number (1.14.99.69), a dedicated Pfam family (PF06175/MiaE) and InterPro entry (IPR010386), and a dedicated primary paper — Carpentier et al. 2020 ([PMID: 32785618](#)) — that reports the in vivo activity, biochemistry, spectroscopy, and crystal structure of MiaE *from P. putida* and explicitly concludes that PP_2188 encodes a functional MiaE. **This step should be marked COVERED with high confidence and no paralog ambiguity.**

However, a species-aware review of the *pathway that supplies MiaE's canonical substrate* uncovers an important curation caveat. The canonical ms2io6A37 route requires an upstream methylthiotransferase, **MiaB (EC 2.8.4.3, K06168)**, to convert i6A37 into the 2-methylthio substrate ms2i6A37 that MiaE then hydroxylates. Three orthogonal lines of evidence — UniProt/InterPro proteome inspection, KEGG orthology, and direct sequence-homology comparison — indicate that **KT2440 has no MiaB**. Its only radical-SAM methylthiotransferase, PP_1197 (Q88NL0), is unambiguously **RimO** (uS12-Asp88 methylthiotransferase, EC 2.8.4.4), not MiaB. A positive/negative control against *P. aeruginosa* PAO1 (which is MiaB-positive and carries two IPR005839 proteins) confirms the KT2440 single-methylthiotransferase count is real, not an annotation gap.

The consequence for module curation is subtle but important. The **enzyme step is covered**, but the **native physiological product in KT2440 is candidate_uncertain**. MiaA (PP_4895, EC 2.5.1.75) is present and produces i6A37, and MiaE is documented to hydroxylate i6A directly in vitro; so in a MiaB-minus background the physiological product is most plausibly **io6A37, not ms2io6A37**. Critically, the Carpentier 2020 demonstration of ms2io6A production was performed in a *heterologous E. coli TOP10 host that supplies ms2i6A* — not in native *P. putida* tRNA — so no published dataset directly establishes that KT2440 makes ms2io6A in vivo. The module's generic "ms2io6A37" product label may therefore be wrong for this organism and should be flagged **module_needs_revision** pending direct LC-MS nucleoside analysis of KT2440 tRNA.

2. Target-Organism Pathway Definition

What UPA00729 includes

UPA00729 covers **exactly one biochemical reaction**: the O₂-dependent, allylic hydroxylation of the isopentenyl side chain of a 2-methylthio-N⁶-isopentenyladenosine at position 37 of tRNA, producing the *cis*-4-hydroxyisopentenyl derivative:

```
ms2i6A37-tRNA + O2 + [reductant] --MiaE--> ms2io6A37-tRNA + H2O
                                     (EC 1.14.99.69)
```

This is the **terminal tail-end modification** in the biosynthesis of the hypermodified wobble-adjacent nucleoside ms2io6A37, found on tRNAs reading codons starting with U (tRNA^{Ser}, tRNA^{Trp}, tRNA^{Tyr}, tRNA^{Phe}, tRNA^{Cys}, etc.). Position 37 (3'-adjacent to the anticodon) modifications stabilize codon-anticodon pairing and maintain reading-frame fidelity.

Neighboring processes to keep separate

- **UPA00729 is a single-gene bucket.** The upstream reactions belong to the same overall biosynthetic *pathway context* but are **not members of this UniPathway bucket**:
- **MiaA** (Δ -isopentenylpyrophosphate:tRNA isopentenyltransferase, EC 2.5.1.75) — makes i6A37 from A37. Present in KT2440 as **PP_4895 (Q88DD2)**.
- **MiaB** (tRNA i6A37 methylthiotransferase, EC 2.8.4.3, K06168) — converts i6A37 → ms2i6A37. **Apparently absent in KT2440** (see §5).

- **RimO** (ribosomal protein uS12 methylthiotransferase, EC 2.8.4.4) is a *paralogous radical-SAM methylthiotransferase acting on protein, not tRNA*. It must be kept strictly separate — it is the source of a real paralog-confusion risk in this proteome (see §5).
- Broad overview maps such as "tRNA modification" (KEGG map / GO:0006400) or "sulfur relay" should not be conflated with this single terminal hydroxylation step.

Alternate names and database definitions

Resource	Identifier	Label
UniPathway	UPA00729	terminal ms2i6A37 → ms2io6A37 hydroxylation
EC	1.14.99.69	tRNA (ms2io6A37) hydroxylase / 2-methylthio-N6-isopentenyladenosine(37)-tRNA monooxygenase
KEGG ortholog	K06169	MiaE
Pfam	PF06175	MiaE
InterPro	IPR010386	tRNA-Hydrxlase_MiaE
GO (molecular function)	GO:0045301	tRNA ms2i6A37 hydroxylase activity
GO (parent function)	GO:0004497	monooxygenase activity
GO (process)	GO:0006400	tRNA modification

Enzyme synonyms in the literature: "MiaE," "tRNA-(ms2io6A37)-hydroxylase," "2-methylthio-N6-isopentenyladenosine(37)-tRNA monooxygenase," "tRNA-modifying monooxygenase."

3. Expected Step Model

The full ms2io6A37 biosynthetic route and its status in KT2440:

```

A37-tRNA
|
| MiaA (EC 2.5.1.75, PP_4895/Q88DD2) [PRESENT]
v
i6A37-tRNA
|
| MiaB (EC 2.8.4.3, K06168) [ABSENT in KT2440]
v
ms2i6A37-tRNA <-- canonical UPA00729 substrate
|
| MiaE (EC 1.14.99.69, PP_2188/Q88KV1) [PRESENT, COVERED]
v
ms2io6A37-tRNA <-- generic module product label

```

Because MiaB is missing but MiaA and MiaE are present, the **most parsimonious native route in KT2440 bypasses the 2-methylthio branch**:

```
A37 --MiaA--> i6A37 --MiaE--> io6A37 (predicted native product)
```

This is mechanistically supported: MiaE is substrate-promiscuous and hydroxylates i6A (yielding io6A), Cl2i6A, and ms2i6A in vitro ([PMID: 23906247](#)).

UPA00729 step status for KT2440:

Step	Enzyme / gene	Status
Terminal hydroxylation (bucket member)	MiaE / PP_2188	covered (direct target-species evidence)
Upstream methylthiolation (context, not in bucket)	MiaB / —	gap / not_present (lineage-specific loss)
Native product identity	—	candidate_uncertain (likely io6A, not ms2io6A)

4. Candidate Genes and Evidence

4.1 MiaE — PP_2188 (Q88KV1) — HIGH CONFIDENCE, COVERED

Role: tRNA (ms2io6A37 / io6A37) hydroxylase; nonheme carboxylate-bridged diiron monooxygenase catalyzing the terminal O₂-dependent allylic hydroxylation.

Evidence type — direct, target-organism, experimental (the strongest possible tier):

- **Reviewed Swiss-Prot, PE1 (protein-level evidence):** Q88KV1 (MIAE_PSEPK), 205 aa, EC 1.14.99.69, Pfam PF06175, InterPro IPR010386. KEGG ppu:PP_2188 = K06169.
- **Four experimental crystal structures:** 2ITB, 6ZMA, 6ZMB, 6ZMC. The enzyme is a diiron four-helix-bundle protein forming a homodimer/monomer with a diiron domain.
- **Dedicated primary paper on the target species:** Carpentier et al. 2020 ([PMID: 32785618](#)) report *"the in vivo activity, biochemical, spectroscopic characterization and X-ray crystal structure of MiaE from Pseudomonas putida"* and state the investigation *"demonstrates that the putative pp-2188 gene encodes a MiaE enzyme."* This is finding **F001**.
- **Mechanistic family assignment:** MiaE contains a *"nonheme dinuclear iron cluster; similar to that found in the hydroxylase component of methane monooxygenase"* (Mathevon et al. 2007, *S. typhimurium*, [PMID: 17679698](#)) — finding **F003**.
- **GO annotations for Q88KV1:** GO:0045301 (tRNA ms2i6A37 hydroxylase activity), GO:0004497 (monooxygenase), GO:0046872 (metal ion binding), GO:0006400 (tRNA modification).

Genomic context (F005): PP_2188 (2,494,792–2,495,409, + strand) is a **standalone transcriptional unit**, flanked by PP_2187 (universal stress protein) and PP_2189 (unknown), both on the complementary strand. There is **no mia operon** — MiaE is not co-transcribed with MiaA or any methylthiotransferase, consistent with its being an independently acquired, lineage-restricted accessory modification enzyme.

Curation-relevant caveat (F006): The Carpentier 2020 in vivo activity was assayed in a **heterologous E. coli TOP10 host**, chosen precisely *because "E. coli contains the unhydroxylated ms2i6A37 in its tRNAs"* — i.e., E. coli supplies the ms2i6A substrate that KT2440 itself may not make. The empty-vector control showed *"ms2i6A37 ... with no evidence for the presence of ms2io6A37,"* while *"tRNAs isolated from the complemented strain showed the presence of ms2io6A37."* **No native P. putida tRNA nucleoside analysis is reported in that paper.** The enzyme's catalytic competence is proven; the identity of its native in vivo product in KT2440 is not.

4.2 Contextual (non-bucket) genes assessed

Gene	Locus / UniProt	Role	Status in KT2440	Evidence
MiaA	PP_4895 / Q88DD2	i6A37 synthase (EC 2.5.1.75)	Present	UniProt + KEGG EC link (F002/F004)
MiaB	—	ms2i6A37 synthase (EC 2.8.4.3, K06168)	Absent	3 orthogonal lines (F002/F004/F007)
RimO	PP_1197 / Q88NL0	uS12-Asp88 methylthiotransferase (EC 2.8.4.4)	Present (paralog, NOT MiaB)	Homology (F004)

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 The MiaB substrate gap (primary finding)

Three independent, orthogonal analyses converge on the conclusion that **KT2440 lacks a canonical MiaB**:

- 1. UniProt/InterPro proteome inspection (F002, F007):** The KT2440 proteome (5,529 proteins) contains **exactly one** IPR005839 (radical-SAM MiaB/RimO fold) protein and **exactly one** MiaE (PF06175). Searches by `gene:miaB`, `ec:2.8.4.3`, `protein_name:methylthiotransferase`, and InterPro IPR005839/IPR013848 return **only RimO** (Q88NL0) — no MiaB.
- 2. KEGG orthology (F002):** Ortholog **K06168 (MiaB)** has **no member in ppv**, whereas it is present in *E. coli* (b0661), *P. aeruginosa* (PA3980), *P. fluorescens* (PFL_5432), and *P. syringae* (PSPTO_4804).
- 3. Direct sequence homology (F004):** 3-mer Jaccard similarity of the sole KT2440 methylthiotransferase PP_1197 (443 aa) is **0.299 to E. coli RimO** (P0AEI4, 441 aa) but only **0.069 to E. coli MiaB** (P0AEI1, 474 aa) — unambiguously identifying PP_1197 as the RimO ortholog (uS12 Asp88, EC 2.8.4.4), not MiaB.

Positive control (F007): *P. aeruginosa* PAO1 (MiaB-positive; PA3980) carries **two** IPR005839 proteins (RimO + MiaB), whereas KT2440 carries only one. This clean positive/negative control demonstrates the KT2440 methylthiotransferase deficit is a genuine biological loss, not a proteome-annotation artifact.

5.2 Native product ambiguity

Because MiaB is absent but MiaA and MiaE are present, the native KT2440 product is most likely **io6A37**, not ms2io6A37. This is supported by MiaE's documented in vitro substrate promiscuity: "*regardless of the substrate (i6A, Cl2i6A, and ms2i6A) used in peroxide-shunt assays, hydroxylation of the terminal isopentenyl-C4-position was observed*" (PMID: 23906247), and the corresponding products io6A, Cl2io6A, and ms2io6A were all synthesized/defined. This is findings **F002/F004**. The generic module product label "ms2io6A37" is therefore likely **wrong for this organism**.

5.3 Paralog / over-annotation risks

- **RimO ↔ MiaB confusion:** PP_1197 is the single greatest annotation-propagation risk in this locus. Both are radical-SAM/IPR005839 methylthiotransferases; naive homology transfer could mislabel it "MiaB." The homology data (F004) firmly assign it to RimO. Curators should ensure PP_1197 is annotated EC 2.8.4.4 (uS12 protein modification), **not** EC 2.8.4.3 (tRNA).
- **Broad EC/GO mappings on MiaE:** Q88KV1 carries the generic GO:0004497 (monooxygenase); the specific, defensible term is GO:0045301 (tRNA ms2i6A37 hydroxylase activity). If native product is io6A, a curator may wish to note that GO:0045301's substrate name (ms2i6A) does not match the likely native substrate (i6A) in this organism — a possible GO term/definition mismatch.
- **Lineage restriction (F003):** MiaE (K06169) is **absent in E. coli**, present in *P. putida*, *P. aeruginosa*, *P. fluorescens*, and *Salmonella*. Kaminska et al. 2008 note ms2i6A is hydroxylated to ms2io6A "*in a few bacteria, such as Salmonella typhimurium*" (PMID: 17910062). Any module that assumes MiaE is broadly distributed across bacteria would be mis-scoped.

6. Module and GO-Curation Recommendations

Module element	Recommended mark	Rationale
MiaE terminal hydroxylation (PP_2188) — enzyme presence	covered	Direct target-species experimental + structural evidence (F001, F005)
Native module product = ms2io6A37	candidate_uncertain / module_needs_revision	MiaB absent → native product most likely io6A, not ms2io6A (F002, F004, F006)
Upstream MiaB substrate step	gap / not_expected_in_target_taxon	Three orthogonal lines confirm MiaB absence; this is a lineage-specific loss (F002, F004, F007)
Upstream MiaA step	present (context only)	PP_4895; not a UPA00729 bucket member

Concrete recommendations:

1. **Mark the UPA00729 enzyme step COVERED.** PP_2188 satisfies the bucket with the highest evidence tier. No paralog ambiguity for MiaE itself.
2. **Flag the module product for revision.** The generic "ms2io6A37" product should carry an organism-specific note: in a MiaB-minus KT2440 background the physiological product is predicted to be **io6A37**. Consider a species-specific module variant or a `candidate_uncertain` annotation on the product node.
3. **Do NOT create a MiaB gap-fill request against PP_1197.** PP_1197 is RimO; annotating it as MiaB would be an error. If module satisfiability logic expects MiaB upstream, mark that step `not_expected_in_target_taxon` rather than forcing a candidate.
4. **Potential GO curation note:** GO:0045301 (tRNA ms2i6A37 hydroxylase activity) is correct for the enzyme's canonical/defined activity but its substrate name may not match the native KT2440 substrate. A broader or additional term capturing i6A→io6A hydroxylation may be warranted if experimental confirmation of io6A is obtained. No new GO term request is urgent absent that data.
5. **Keep RimO and the tRNA hydroxylation module strictly separate** in the module graph to prevent cross-propagation.

7. Genes to Promote to Full Review

Gene	Promote?	Reason
miaE / PP_2188 (Q88KV1)	Yes — confirm/close	Already the bucket's sole member with PE1 + structures; a full <code>fetch-gene</code> review will formally close it as COVERED and attach the substrate-supply caveat.
PP_1197 (Q88NL0, RimO)	Yes — disambiguation	Full review needed to lock its annotation as RimO (EC 2.8.4.4) and pre-empt any MiaB mis-propagation in the module graph.
PP_4895 (Q88DD2, MiaA)	Optional	Context gene; confirms i6A37 production. Low priority — not a bucket member.

8. Limitations and Knowledge Gaps

- **No native KT2440 tRNA nucleoside data.** The single most important gap: every ms2io6A demonstration for Pp-MiaE was performed in *heterologous E. coli* (F006). Whether KT2440 tRNA carries io6A, ms2io6A, both, or neither is **experimentally unresolved**.
- **MiaB absence is inferred from bioinformatics**, albeit by three orthogonal methods with a clean positive control. A cryptic, highly diverged methylthiotransferase cannot be formally excluded, though InterPro/KEGG/homology all agree it is unlikely.
- **Homology metric is coarse.** The RimO-vs-MiaB assignment used a 3-mer Jaccard heuristic (0.299 vs 0.069); this is decisive directionally but is not a substitute for a full phylogenetic/HMM analysis.
- **Substrate promiscuity is in vitro.** MiaE's ability to hydroxylate i6A directly ([PMID: 23906247](#)) was shown with nucleoside surrogates in peroxide-shunt assays, not on native tRNA in KT2440.

9. Proposed Follow-up Experiments / Actions

1. **LC-MS/MS nucleoside analysis of KT2440 total tRNA** (highest priority). Directly determine whether io6A37 and/or ms2io6A37 is present. This single experiment resolves the product-identity ambiguity and the module_needs_revision flag.
2. **ΔPP_2188 knockout + LC-MS**. Confirm that the observed hydroxylated species (io6A or ms2io6A) disappears upon miaE deletion, tying the modification to PP_2188 in vivo.
3. **Bioinformatic confirmation of MiaB absence** via HMMER/pHMM search of the KT2440 proteome and re-annotation of PP_1197 as RimO with a curator note.
4. **Cross-species check within the KT2440 lineage**. Confirm whether other sequenced *P. putida* strains share the MiaB-minus/MiaE-plus configuration, to establish whether this is strain- or species-level.
5. **Expert question for module curators**: Should UPA00729's product node be organism-parameterized (io6A vs ms2io6A) rather than fixed, given MiaE's substrate promiscuity and variable upstream MiaB presence across taxa?

10. Key References

PMID	Title (abbrev.)	Support for this review
32785618	<i>Structural, biochemical and functional analyses of tRNA-monooxygenase MiaE from Pseudomonas putida</i>	Primary target-organism evidence that PP_2188 encodes functional MiaE (F001); also reveals the in vivo assay used heterologous <i>E. coli</i> (F006).
17679698	<i>tRNA-modifying MiaE from Salmonella typhimurium is a nonheme diiron monooxygenase</i>	Defines the enzyme family/mechanism (nonheme diiron, MMO-like) underlying EC 1.14.99.69 (F003).
23906247	<i>Peroxide-shunt substrate-specificity for the S. typhimurium MiaE</i>	MiaE hydroxylates i6A, Cl2i6A, and ms2i6A → io6A, Cl2io6A, ms2io6A; basis for predicting io6A as native KT2440 product (F002, F004).
17910062	<i>Structural bioinformatics of ms2io6A37 biosynthesis enzymes</i>	Establishes taxonomic restriction of the ms2io6A step to "a few bacteria such as Salmonella typhimurium" (F003).

Bottom line for curators

Mark the **UPA00729 MiaE hydroxylation step COVERED** (PP_2188, direct + structural evidence). Flag the **module product for revision** because the upstream MiaB methylthiotransferase is absent in KT2440 (three orthogonal lines + positive control), making the native product most plausibly **io6A37 rather than ms2io6A37** — a conclusion resolvable only by direct LC-MS of KT2440 tRNA. Do not mis-annotate PP_1197 as MiaB; it is RimO.

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